

Now it remains to choose particular values of $\sigma, n, p, q, \{r_i\}, \gamma$ such that $\gamma^2 \geq \frac{1}{n}$ and $\text{MMD}^2(\eta_{r^*}, \kappa_{r^*}; k) \geq \text{MMD}^2(p, q; k)$. In fact, the values chosen for Case 1 as in Table 2 already yields the previous inequalities for Case 2. $\square$

A.4. Proofs of Lemma 1 and Proposition 2

Proof of Lemma 1

Proof. For all n , we have

$$\frac{|a_n|}{f(n)} \leq \frac{|X_n|}{f(n)} \text{ a.s.} \quad (5)$$

Denote by P the probability measure of the underlying measurable space defining the random variables X_n . Since $X_n = O_p(f(n))$, for any $\epsilon > 0$, there exists $M_\epsilon > 0, N_\epsilon > 0$ such that

$$P\left(\frac{|X_n|}{f(n)} > M_\epsilon\right) \leq \epsilon, \forall n > N_\epsilon.$$

It follows from Eq. (5) that $\forall n, \left\{\frac{|a_n|}{f(n)} \leq M_\epsilon\right\} \supseteq \left\{\frac{|X_n|}{f(n)} \leq M_\epsilon\right\}$ which implies that for all $n > N_\epsilon$, we have

$$\begin{aligned} 1_{\left\{\frac{|a_n|}{f(n)} \leq M_\epsilon\right\}} &= P\left(\frac{|a_n|}{f(n)} \leq M_\epsilon\right) \\ &\geq P\left(\frac{|X_n|}{f(n)} \leq M_\epsilon\right) \\ &\geq 1 - \epsilon, \end{aligned}$$

where the equality is due to that a_n are deterministic. Picking any $\epsilon \in (0, 1)$, we have

$$\frac{|a_n|}{f(n)} \leq M_\epsilon, \forall n > N_\epsilon,$$

which implies $a_n = O(f(n))$. $\square$

Proof of Proposition 2 First, we state the following proposition.

Proposition 3. Assume that $\sup_{x,y} k(x,y) \leq B$. Let $X_1, \dots, X_n$ be n i.i.d. samples of P and denote $\xi(X_1, \dots, X_n) := \text{MMD}\left(\frac{1}{n} \sum_{i=1}^n \delta_{X_i}, P; \mathcal{H}\right)$. For any $\epsilon > 0$, we have

$$P\left(\xi(X_1, \dots, X_n) > \epsilon + 2\sqrt{\frac{B}{n}}\right) \leq \exp\left(-\frac{\epsilon^2 n}{2B}\right).$$

Note that by setting $t = \exp\left(-\frac{\epsilon^2 n}{2B}\right)$ in Proposition 3, we have

$$P\left(\xi(X_1, \dots, X_n)/(1/\sqrt{n}) > M_t\right) \leq t,$$

where $M_t := \sqrt{2B \log(1/t)} + 2\sqrt{B}$. Thus, $\xi(X_1, \dots, X_n) = O_p(1/\sqrt{n})$, which concludes Proposition 2.

Now, we only need to prove Proposition 3. The proof follows a standard procedure to bound an empirical process $\xi(X_1, \dots, X_n)$ where we first bound it in probability w.r.t. its expectation using concentration inequalities and then we bound its expectation via a complexity notation of the witness function class $\mathcal{H}$.

Preliminaries. Before proving Proposition 3, we present some relevant notations and preliminary results from which we combine to derive a proof for Proposition 3. For any function $g : \mathcal{X}^n \rightarrow \mathbb{R}$, denote

$$\begin{aligned} \delta_i g(x_1, \dots, x_n) &:= \sup_z g(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_n) - \inf_z g(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_n), \\ \|\delta_i g\|_\infty &:= \sup_{x_1, \dots, x_n} |\delta_i g(x_1, \dots, x_n)|. \end{aligned}$$

Denote by $\mathcal{H} := \{f \in \mathcal{F} : \|f\|_{\mathcal{F}} \leq 1\}$ the unit ball of the RKHS $\mathcal{F}$. For any $\{x_i\}_{i=1}^n \in \mathcal{X}^n$, we denote $\mathcal{H} \circ \{x_1, \dots, x_n\} := \{(f(x_1), \dots, f(x_n)) \in \mathbb{R}^n : f \in \mathcal{H}\}$.

Lemma 7 (McDiarmid's inequality (Rebeschini 2019)). Let $X_1, \dots, X_n$ be independent random variables in $\mathcal{X}$. Let $g : \mathcal{X}^n \rightarrow \mathbb{R}$ be any function such that $\|\delta_i g\|_\infty < \infty, \forall i$. For any $\epsilon > 0$, we have

$$P(g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n) \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n \|\delta_i g\|_\infty^2}\right).$$

Definition 2. The Rademacher complexity of a set $\mathcal{T} \subseteq \mathbb{R}^n$ is defined as

$$\text{Rad}(\mathcal{T}) := \mathbb{E} \sup_{t \in \mathcal{T}} \frac{1}{n} \sum_{i=1}^n \Omega_i t_i,$$

where $\Omega_1, \dots, \Omega_n \in \{-1, 1\}$ are independent Rademacher random variables, i.e., $P(\Omega_i = 1) = P(\Omega_i = -1) = 1/2, \forall i$.

Lemma 8. We have

$$\mathbb{E}[\xi(X_1, \dots, X_n)] \leq 2\mathbb{E}\text{Rad}(\mathcal{H} \circ \{X_1, \dots, X_n\}).$$

In addition, we have

$$\mathbb{E}\text{Rad}(\mathcal{H} \circ \{X_1, \dots, X_n\}) \leq \sqrt{\frac{B}{n}}.$$

Proof. Let $\tilde{X}_1, \dots, \tilde{X}_n$ be new independent samples from P and independent of $S := \{X_1, \dots, X_n\}$. We have

$$\begin{aligned} \mathbb{E}[\xi(X_1, \dots, X_n)] &= \mathbb{E}\left[\text{MMD}\left(\frac{1}{n} \sum_{i=1}^n \delta_{X_i}, P; \mathcal{F}\right)\right] = \mathbb{E} \sup_{f \in \mathcal{H}} \left(\mathbb{E}f(X) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right) \\ &= \mathbb{E} \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \mathbb{E} \left[f(\tilde{X}_i) - f(X_i) \middle| S \right] \\ &\stackrel{(a)}{\leq} \mathbb{E} \mathbb{E} \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (f(\tilde{X}_i) - f(X_i)) \\ &= \mathbb{E} \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (f(\tilde{X}_i) - f(X_i)) \\ &\stackrel{(b)}{=} \mathbb{E} \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \Omega_i (f(\tilde{X}_i) - f(X_i)) \\ &\leq \mathbb{E} \left[\sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \Omega_i f(\tilde{X}_i) + \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (-\Omega_i) f(X_i) \right] \\ &= 2\mathbb{E} \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \Omega_i f(X_i) \\ &= 2\mathbb{E}\text{Rad}(\mathcal{H} \circ \{X_1, \dots, X_n\}). \end{aligned}$$

Here (a) follows from Jensen's inequality for convex function $\sup$ and (b) follows from $(f(\tilde{X}_i) - f(X_i))_{1 \leq i \leq n}$ has the same distribution as $(\Omega_i(f(\tilde{X}_i) - f(X_i)))_{1 \leq i \leq n}$.

In addition, we have

$$\begin{aligned} \mathbb{E}\text{Rad}(\mathcal{H} \circ \{X_1, \dots, X_n\}) &= \mathbb{E} \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \Omega_i f(X_i) \\ &= \mathbb{E} \sup_{f \in \mathcal{H}} \left\langle f, \frac{1}{n} \sum_{i=1}^n \Omega_i k(X_i, \cdot) \right\rangle_{\mathcal{F}} \\ &\leq \mathbb{E} \sup_{f \in \mathcal{H}} \|f\|_{\mathcal{F}} \times \left\| \frac{1}{n} \sum_{i=1}^n \Omega_i k(X_i, \cdot) \right\|_{\mathcal{F}} \quad (\text{Cauchy-Schwartz inequality}) \end{aligned}$$

$$\begin{aligned}
&\leq \mathbb{E} \left\| \frac{1}{n} \sum_{i=1}^n \Omega_i k(X_i, \cdot) \right\|_{\mathcal{F}} \\
&= \frac{1}{n} \mathbb{E} \sqrt{\sum_{i,j} \Omega_i \Omega_j k(X_i, X_j)} \\
&\leq \frac{1}{n} \sqrt{\mathbb{E} \sum_{i,j} \Omega_i \Omega_j k(X_i, X_j)} \text{ (Jensen's inequality)} \\
&= \frac{1}{n} \sqrt{\mathbb{E} \sum_{i=1}^n k(X_i, X_i)} \leq \frac{1}{n} \sqrt{nB} = \sqrt{\frac{B}{n}}.
\end{aligned}$$

The last equality is due to that $\Omega_i^2 = 1, \forall i$ and $\mathbb{E} [\Omega_i \Omega_j] = 0, \forall i \neq j$. $\square$

Proof of Proposition 3. Now, we are ready to prove Proposition 3. For any $\{x_i\}_{i=1}^n \in \mathcal{X}^n$, we have

$$\begin{aligned}
\delta_i \xi(x_1, \dots, x_n) &\leq \sup_z \left\| \frac{1}{n} \sum_{j=1, j \neq i}^n k(x_j, \cdot) + \frac{k(z, \cdot)}{n} - \int k(x, \cdot) dP(x) \right\|_{\mathcal{F}} \\
&\quad - \inf_z \left\| \frac{1}{n} \sum_{j=1, j \neq i}^n k(x_j, \cdot) + \frac{k(z, \cdot)}{n} - \int k(x, \cdot) dP(x) \right\|_{\mathcal{F}} \\
&\leq \sup_{z, z'} \left\| \frac{k(z, \cdot)}{n} - \frac{k(z', \cdot)}{n} \right\|_{\mathcal{F}} \leq \frac{2\sqrt{B}}{n}.
\end{aligned}$$

Thus, it follows from McDiarmid's inequality that

$$P(\xi(X_1, \dots, X_n) - \mathbb{E}\xi(X_1, \dots, X_n) \geq \epsilon) \leq \exp\left(-\frac{2n\epsilon^2}{B}\right). \quad (6)$$

Eq. (6) and Lemma 8 immediately yield Proposition 3. $\square$

Appendix B. Algorithm details

In this appendix, we present the algorithm details for the tabular and Atari experiments in the main text.

Tabular experiment

In tabular case, the particles $Z_\theta(s, a)$ reduces to tabular representation $\{\theta_i(s, a)\}_{i=1}^N$, so a return distribution is represented as a mixture of Diracs

$$\mu(s, a) = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i(s, a)}.$$

The algorithm details used for tabular policy evaluation in the main text is presented in Algorithm 2. In both tabular MMD-DRL and QRDL cases, the TD gradients have a closed-form expression which we explicitly used in our tabular experiment. The detailed values of each (hyper-)parameters of the algorithms used in our experiment are reported in Table 3.

(Hyper-)Parameters	Values
Learning rate schedule	$\alpha_t = \frac{1}{t^{0.2}}$
Particle initialization	$\mathcal{N}(-1, 0.08)$
Number of episodes per iteration	100
Number of iterations	15
Number of particles N	30
Number of MC rollouts	10,000
Kernel bandwidth h (MMDRL only)	$\{8, 10, 12\}$
Quantiles (QRDL only)	$\{\frac{2i-1}{2N} : 1 \leq i \leq N\}$

Table 3: The (hyper-)parameters of the algorithms used in our tabular policy evaluation experiment with the Chain MDP.